

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.4: Surface Area and Volume of Solids — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.4: Surface Area and Volume of Solids
Driving Question	DRIVING QUESTION: How does the shape of a solid determine how much material is needed to build it and how much it can hold — and how do engineers and manufacturers in Kenya use this knowledge to design efficient, cost-effective structures?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Surface Area of Prisms and Pyramids — Launching the Phenomenon	Learners examined physical or drawn nets of rectangular prisms (e.g., a cereal box or brick), triangular prisms, and square-based pyramids. They unfolded (or sketched) each net, labelled every face, and measured/recorded the dimensions of each face. They observed that every face of a prism is a rectangle or triangle, that opposite faces of a rectangular prism are congruent, and that a square pyramid has one square base and four congruent triangular lateral faces. They also observed that the total surface area equals the combined area of all faces in the net.	Learners derived and applied three formulae: (1) Rectangular prism: $SA = 2(lw + lh + wh)$; (2) Triangular prism: $SA = bh_{\text{triangle}} + (a + b + c)l$, where the first term is twice the triangular base area and the second term is the lateral area; (3) Square pyramid: $SA = b^2 + 2bl_{\text{slant}}$, where b is the base side length and l_{slant} is the slant height of each triangular face. They learned that slant height $\neq$ vertical height, and that the slant height is found using the Pythagorean theorem: $l = \sqrt{(h^2 + (b/2)^2)}$. TEACHER NOTE: Emphasise the distinction between slant height and vertical height here — it recurs in every subsequent lesson on cones and frustums.	The shape of a solid determines its surface area because each face contributes a calculable area to the total. For the anchoring phenomenon (designing Kenyan packaging and buildings), a rectangular prism-shaped maize flour bag requires material equal to $2(lw + lh + wh)$; reducing any dimension directly reduces material cost. A square-pyramid roof over a square room requires sheeting equal to $b^2 + 2bl_{\text{slant}}$ — so a steeper roof (larger slant height) demands more roofing iron sheets. Learners explained why a manufacturer in Eldoret would calculate surface area before ordering sheet metal or cardboard: to avoid waste and control cost.
Lesson 2 Surface Area of Cones and Spheres	Learners observed a physical cone (e.g., a paper ice-cream cone or a party hat) and a sphere (e.g., a football or orange). They cut and unrolled the cone's lateral surface to reveal a sector of a circle, noting that the arc length of the sector equals the circumference of the cone's base. For the sphere, they observed a demonstration (or video) in which an orange peel was	Learners learned two key formulae: (1) Total surface area of a cone: $SA = \pi r^2 + \pi rl = \pi r(r + l)$, where r is the base radius and l is the slant height — NOT the vertical height. (2) Surface area of a sphere: $SA = 4\pi r^2$. They also derived $l = \sqrt{(h^2 + r^2)}$. TEACHER NOTE: A common error is substituting the vertical height h in place of slant height l .	A conical roof on a Kenyan banda (thatched hut) or a conical metal funnel requires material equal to πrl for the curved surface, plus πr^2 if the base is to be covered. A spherical water storage tank (e.g., elevated tanks seen along Mombasa Road) requires painting material equal to $4\pi r^2$. Because the sphere's surface area grows as r^2 , doubling the radius quadruples the paint required —

	flattened to show it covers approximately four times the area of a circle with the same radius. They recorded: cone has one circular base and one curved lateral face; sphere has no flat faces.	Anticipate this misconception explicitly — ask learners to identify both h and l on a diagram before any calculation.	a critical cost-planning insight for manufacturers and civil engineers. This lesson directly advances the driving question by connecting shape geometry to material quantity.
Lesson 3 Surface Area of Frustums of a Cone	Learners observed a frustum of a cone — a truncated cone — created by horizontally slicing the top off a conical model (or examining a bucket/water tank). They identified the two circular faces (large base radius R and small top radius r) and the curved lateral surface. They observed that the lateral surface, when unrolled, forms a sector of an annulus (a strip of a circular ring), and recorded that the frustum has three distinct surface regions: bottom circle, top circle, and curved lateral band.	Learners derived and applied the total surface area formula for a frustum of a cone: $SA = \pi R^2 + \pi r^2 + \pi(R + r)l$, where l is the slant height of the frustum given by $l = \sqrt{(h^2 + (R - r)^2)}$. The derivation proceeds by computing the lateral surface area of the large cone minus the lateral surface area of the removed small cone, using similar-triangle proportions to relate their slant heights. TEACHER NOTE: Walk learners through the similar-triangle derivation step by step — the proportional reasoning used here ($R/r = L/l_{\text{small}}$) is a high-value skill that reappears in Lesson 7 (volume of frustums of a cone).	Conical frustums are extremely common in Kenyan daily life: metal buckets, plastic basins, grain storage silos, and the frustum-shaped charcoal jikos (stoves) all approximate frustum geometry. Calculating $SA = \pi R^2 + \pi r^2 + \pi(R + r)l$ tells a metalworker in Kamuthe market exactly how much sheet metal to cut for a jiko or bucket, minimising waste. Learners explained that the difference between R and r (how 'tapered' the frustum is) directly affects the lateral surface area, connecting shape design to material efficiency.
Lesson 4 Surface Area of Frustums of a Pyramid	Learners observed a physical or drawn model of a frustum of a square pyramid (a truncated pyramid with a larger square base of side a and a smaller square top of side b). They unfolded its net and identified: one large square base (area a^2), one small square top (area b^2), and four congruent trapezoidal lateral faces. They measured and recorded all dimensions, noting that each trapezoidal face has parallel sides a and b and a slant height l_f .	Learners derived the surface area formula for a frustum of a square pyramid: $SA = a^2 + b^2 + 2l_f(a + b)$, where l_f is the slant height of each trapezoidal face, found using $l_f = \sqrt{(h^2 + ((a - b)/2)^2)}$. The derivation proceeds by decomposing the net: two square faces + four trapezoids. Area of one trapezoid = $\frac{1}{2}(a + b) \times l_f$, so four trapezoids contribute $4 \times \frac{1}{2}(a + b) \times l_f = 2l_f(a + b)$. TEACHER NOTE: Insist learners sketch and label the net before writing any formula — spatial decomposition is the core skill. Confirm that learners are computing l_f correctly and not confusing it with the pyramid's vertical height.	Frustums of pyramids appear in real Kenyan construction contexts: the truncated pyramid shape is used in some traditional granary roofs, decorative architectural elements, and in engineering structures. A construction engineer calculating the amount of roofing or cladding material for a frustum-shaped pavilion top in Nairobi's Uhuru Park would use $SA = a^2 + b^2 + 2l_f(a + b)$. Learners explained that increasing the taper (making a and b more different) increases the lateral surface area, meaning more material is needed — a direct answer to the driving question.
Lesson 5 Surface Area of Composite Solids	Learners observed a composite solid — a cylinder topped with a hemisphere (modelling a common Kenyan water storage tank, as seen in rural homes and schools). They identified which surfaces are exposed (the cylinder's lateral surface, the cylinder's bottom circular base, and the hemisphere's curved surface) and which surface is internal/hidden (the flat circle shared between the cylinder top and	Learners learned the principle: SA of a composite solid = sum of the surface areas of component parts – $2 \times$ area of each shared/hidden face. For the cylinder-hemisphere tank with cylinder radius r and height h_c : $SA = (2\pi r \cdot h_c + \pi r^2) + (2\pi r^2) - \pi r^2 = 2\pi r h_c + 2\pi r^2$. Substituting $r = 0.9$ m and $h_c = 1.0$ m gives $SA \approx 5.12\pi \approx 16.09$ m ² . TEACHER NOTE: The most common error is either adding the shared face twice	Real Kenyan water tanks (e.g., Polytank or fabricated steel tanks installed by NGOs in Turkana County) often combine cylindrical and hemispherical sections for structural strength and volume efficiency. A technician ordering fibreglass or galvanised steel sheet for such a tank must calculate only the exposed surface area — adding the hemisphere's curve and the cylinder's body while excluding the internal join. This lesson

	hemisphere base). They recorded all relevant dimensions and noted that the shared face must be excluded from the surface area calculation.	or forgetting to subtract it at all. Use a colour-coded diagram to make visible which faces are 'open to the outside' versus 'sealed inside.' Reinforce that only exposed surfaces require material.	completes the surface area block of the driving question, showing that even complex real-world shapes are solved by decomposition into known solids.
Lesson 6 Volume of Prisms, Pyramids, Cones, and Spheres	Learners observed a hands-on or demonstrated filling experiment: a conical container and a cylindrical container of equal base radius and equal height were filled with sand or water. It took exactly three full cones to fill the cylinder, establishing $V_{\text{cone}} = \frac{1}{3} V_{\text{cylinder}}$. Similarly, a pyramid-to-prism filling experiment (or video demonstration) confirmed $V_{\text{pyramid}} = \frac{1}{3} V_{\text{prism}}$. Learners recorded all volumes and dimensions and noted the 1:3 relationship between pyramid/cone and their corresponding prism/cylinder.	Learners learned and applied four volume formulae: (1) Prism/Cylinder: $V = A_{\text{base}} \times h$ (for cylinders: $V = \pi r^2 h$); (2) Pyramid/Cone: $V = \frac{1}{3} \times A_{\text{base}} \times h$ (for cones: $V = \frac{1}{3} \pi r^2 h$); (3) Sphere: $V = \frac{4}{3} \pi r^3$. They also practised converting between units (cm^3 , m^3 , litres: 1 litre = 1000 cm^3). TEACHER NOTE: Emphasise the $\frac{1}{3}$ factor conceptually — it is not arbitrary but emerges from the filling experiment. Ask learners: 'What would happen to the volume of a cone if you doubled the radius? Doubled the height?' (Doubling r quadruples volume; doubling h doubles it.) This builds proportional intuition for Lessons 7 and 8.	Volume answers 'how much can it hold' — the second half of the driving question. A cylindrical water storage tank in Kisumu with $r = 1.5$ m and $h = 2$ m holds $V = \pi(1.5)^2(2) \approx 14.14 \text{ m}^3 = 14,140$ litres. A conical grain storage bin holds $\frac{1}{3}$ as much as a same-sized cylinder — relevant for a maize farmer in the Rift Valley choosing between bin shapes. Learners explained that the formula choice depends entirely on the base shape and whether the solid is a prism-type or pyramid-type, directly linking geometry to practical holding capacity.
Lesson 7 Volume of Frustums of a Cone	Learners observed that a frustum of a cone can be reconstructed into a full cone by extending its sides to meet at an apex. They used similar-triangle proportions (with radii R and r and the frustum height h) to locate the apex and determine the heights of the large cone (H) and the small removed cone ($H - h$). They recorded the relationship $R/r = H/(H - h)$ and confirmed it with a physical model or diagram. They observed that $V_{\text{frustum}} = V_{\text{large cone}} - V_{\text{small cone}}$.	Learners derived and applied the volume formula for a frustum of a cone: $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$, obtained by computing $V_{\text{large}} - V_{\text{small}}$ and simplifying algebraically using the proportional relationship between R , r , and H . They verified the formula with numerical examples (e.g., a frustum with $R = 6$ cm, $r = 3$ cm, $h = 4$ cm gives $V = \frac{1}{3} \pi (4)(36 + 18 + 9) = 84\pi \approx 263.9 \text{ cm}^3$). TEACHER NOTE: Work through the algebraic derivation on the board in full — it is an excellent opportunity to reinforce algebraic manipulation skills. Many learners will need two passes through the similar-triangle proportion step before it is secure.	Frustum-of-cone volumes are essential for designing and calibrating containers with tapered profiles: water buckets, conical grain storage tanks, and industrial hoppers used in Kenyan food-processing factories (e.g., grain mills in Nakuru). A factory engineer designing a hopper that must hold exactly 500 litres of maize flour uses this formula to determine the required dimensions. Learners explained that the presence of both R^2 and Rr and r^2 in the formula means all three cross-section dimensions (both radii and the height) interact to determine capacity — no dimension can be ignored.
Lesson 8 Volume of Frustums of a Pyramid	Learners observed a frustum of a square pyramid (truncated pyramid) and decomposed it mentally into: (i) the large original pyramid, and (ii) the small pyramid removed from the top. Using similar-shape reasoning, they established the relationship between the base sides a , b , and the heights H and $(H - h)$. They recorded $A_1 = a^2$ (area of large base) and $A_2 = b^2$ (area of	Learners derived and applied the volume formula for a frustum of a pyramid: $V = \frac{h}{3} (A_1 + A_2 + \sqrt{A_1 A_2})$, where A_1 and A_2 are the areas of the two parallel bases and h is the perpendicular height of the frustum. This formula is general — it applies to frustums of any pyramid (square, rectangular, or triangular base) provided A_1 and A_2 are computed correctly. TEACHER NOTE:	Frustums of pyramids appear in large-scale Kenyan engineering contexts: the earthwork embankments of road cuttings (common along the Nairobi–Nakuru Highway), retaining walls with tapered cross-sections, and the traditional Nubian architecture influenced by ancient pyramidal forms. A civil engineer calculating the volume of soil to be excavated for a tapered foundation pit

	small top) and confirmed that $\sqrt{(A_1A_2)} = ab$ (the geometric mean of the two base areas).	Connect this formula explicitly to the cone frustum formula from Lesson 7 — they share the same structure. The term $\sqrt{(A_1A_2)}$ is the geometric mean base area, analogous to the Rr term in the cone frustum formula.	uses $V = (h/3)(A_1 + A_2 + \sqrt{(A_1A_2)})$. Learners explained that the geometric mean term $\sqrt{(A_1A_2)}$ accounts for how the cross-section changes continuously between the two bases, not just abruptly.
Lesson 9 Volume of Composite Solids — Solving the Kisumu Tank Mystery	Learners revisited the cylinder-hemisphere composite solid from Lesson 5 (now computing its volume, not its surface area). They identified the two component solids — a full cylinder of radius r and height h_c , and a hemisphere of radius r — and noted that the hemisphere sits on top of the cylinder. They observed that the total volume is simply the sum of the two parts, with no subtraction needed (unlike surface area, where shared faces must be removed). They recorded: $V_{\text{cylinder}} = \pi r^2 h_c$ and $V_{\text{hemisphere}} = (2/3)\pi r^3$.	Learners learned that: (a) $V_{\text{composite}} =$ sum of component volumes (no subtraction for internal joins, unlike surface area); (b) $V_{\text{cylinder-hemisphere}} = \pi r^2 h_c + (2/3)\pi r^3$; (c) substituting $r = 0.9$ m and $h_c = 1.0$ m gives $V \approx 2.545 + 1.527 \approx 4.07 \text{ m}^3 \approx 4,070$ litres. They also practised computing volumes of other composite solids (e.g., a cone on top of a cylinder, a prism with a pyramidal roof). TEACHER NOTE: Explicitly contrast with Lesson 5 — surface area requires subtracting hidden faces; volume requires only addition. This conceptual distinction is a frequent exam trap.	The Kisumu Tank Mystery is resolved: the community tank holds approximately 4,070 litres. This volume calculation tells the water project engineer how many households can be served per refill. Learners explained that choosing a cylinder-hemisphere shape (rather than a pure cylinder) gives a slightly larger volume for the same height, because the dome adds volume above the cylinder. This lesson ties together the entire volume block and answers the 'how much can it hold' component of the driving question using a real Kenyan water-access scenario.
Lesson 10 Real-Life Applications, Final Explanation and Cumulative Model Presentation	Learners examined two real Kenyan design scenarios side by side: (1) a cylindrical and a rectangular packaging box with identical volumes — and compared their surface areas; (2) a spherical water tank and a cubic water tank of equal volume — and compared their surface areas. They observed from calculations that the sphere requires less surface material than the cube for the same volume, and that a cylinder uses less material than a rectangular box for equivalent volume. Learners also reviewed all their model revisions from Lessons 1–9 and consolidated them into a single cumulative diagram.	Learners learned that surface area and volume are independent properties — a solid's shape can be redesigned to preserve volume while reducing surface area (material efficiency) or to preserve surface area while increasing volume (capacity efficiency). Key principle: among all solids of a given volume, the sphere has the minimum surface area. They also reviewed all formulae from Lessons 1–9 in an organised reference table. TEACHER NOTE: Use this lesson to explicitly revisit every formula derived across the sub-strand — a formula consolidation table on the board (with derivation cues, not just final forms) is highly effective. Allow learners to self-correct their individual formula sheets.	The driving question is fully answered: shape determines both how much material is needed (surface area) and how much a solid can hold (volume). Kenyan engineers and manufacturers — from Bamburi Cement designing clinker storage silos, to Bidco packaging cooking oil, to water engineers in Marsabit designing community tanks — optimise shape to minimise cost and maximise capacity. A sphere minimises material for a given volume; a frustum-shaped bucket balances stackability, strength, and capacity. Learners presented their cumulative models showing how each lesson added one layer of understanding toward this complete, quantitative answer.